Study of the out-of-equilibrium dynamics of one-dimensional Bose gases

Étude de la dynamique hors équilibre des gaz de Bosons unidimensionnels

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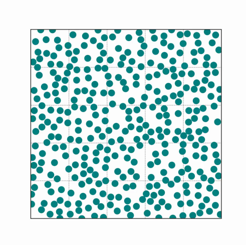
github.com/THEMEZE

Rapidities and GGE in a classical system

Chaotic/Ergodic

Integrable/non-ergidic



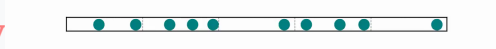
particule density

n

nu ← velocity

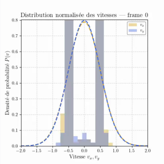
ε

energie density

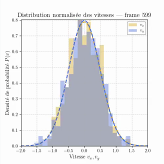


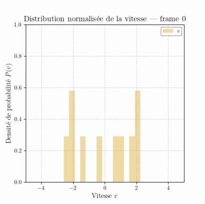
$\rho(v)$

velocity density



thermalisation $\Longrightarrow$





Thesis Problematic

Scientific Question

How can we characterize the spatio-temporal evolution of a 1D quantum gas after a perturbation, in terms of the local rapidity distribution and its link to Generalized Gibbs Ensembles (GGE)?

Objectives

- Lieb and Liniger;
- GHD;
- Manip;
- Bipartition;
- Fluctuation

The N-body problem: the scattering phase

Kinetic energy

Contact potential

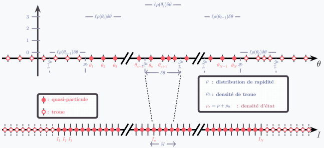
$$\hat{H} = -\frac{\hbar^2}{2m}\sum_{a=1}^{N} \hat{\partial}_{x_a}^2 + g \sum_{1\leqslant a<b\leqslant N} \delta(\hat{x}_a - \hat{x}_b)$$

1D repulsion strength

$\{\theta_1, \cdots, \theta_N\}$

$$\psi_{\{\theta_a\}}(x_1 < \cdots < x_N) \propto \sum_{\text{perm. }\sigma} e^{i\Phi\,\sigma(\{\theta_a\})}\, e^{i\frac{m}{\hbar}\sum_{b=1}^{N}\theta_{\sigma(b)} x_b}$$

$|\{\theta_a\}\rangle$



$$\frac{m}{\hbar}\theta_a L + \sum_{b=1}^{N}\Phi(\theta_a - \theta_b) = 2\pi I_a \quad a = 1, \cdots, N$$

$$\begin{cases} L\rho(\theta)\delta\theta & : \# \quad \text{rapidities} \quad \in [\theta, \theta+\delta\theta] \\ L\rho_h(\theta)\delta\theta & : \# \quad \text{holes} \quad \in [\theta, \theta+\delta\theta] \\ L\rho_s(\theta)\delta\theta & : \# \quad \text{states} \quad \in [\theta, \theta+\delta\theta] \end{cases}$$

The N-body problem: the scattering phase

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Contact potential

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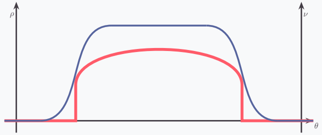

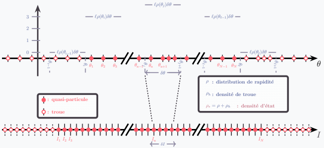

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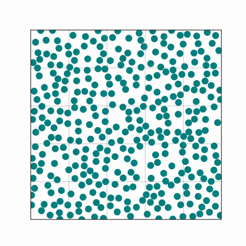
GHD

Hydrodynamic scale

Chaotic/Ergodic

Integrable/non-ergidic



local particle density

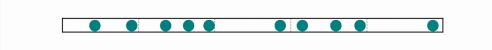
$$q_n(\boldsymbol{r};t) = n(\boldsymbol{r};t)$$

local momentum density

$$q_p(\boldsymbol{r};t) = nu(\boldsymbol{r};t)$$

$$q_e(\boldsymbol{r};t) = e(\boldsymbol{r};t)$$

local energie density



$$\rho(x, \theta; t)$$

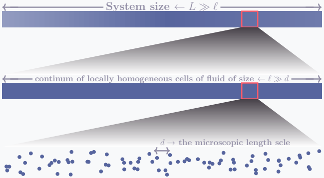
local velocity density

Hydrodynamic approach:

- Large-scale dynamics.
- The spatio-temporal space is decomposed into cells of size $\tau \times \ell$.

Concept of local relaxation

in an isolated quantum system



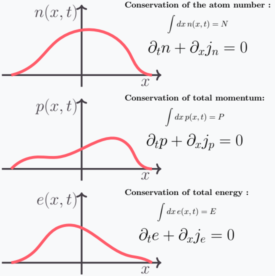
GHD

Hydrodynamic scale

Chaotic/Ergodic

Hydrodynamics for a non-integrable quantum system:

Locally described by a Gibbs ensemble with $n(x,t)$, $p(x,t)$ and $e(x,t)$.



$$\partial_t q_i + \partial_x j_i = 0, \quad i \in \{n, p, e\}$$

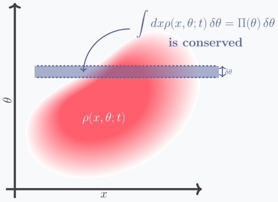
At the Euler scale:

$$j_i(x;t) \simeq f(n(x;t), p(x;t), e(x;t))$$

Integrable/non-ergidic

Generalized hydrodynamics for an integrable quantum system:

Local description by a Generalized Gibbs Ensemble parameterized by the local rapidity distribution $\rho(\theta, x, t)$.



$$\partial_t \rho(\theta) + \partial_x j(\theta) = 0, \quad \theta \in \mathbb{R}$$

At the Euler scale:

$$j(x,\theta;t) \simeq f(\rho(x,\cdot;t))$$

A frame with title and subtitle

Subtitle here

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Itemized list:

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  - Consectetur
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- Sed do eiusmod
  - Tempor incididunt
    - Ut labore et dolore
    - Magna aliqua

A frame with title only

Theorem

$$e^{i\pi} + 1 = 0$$

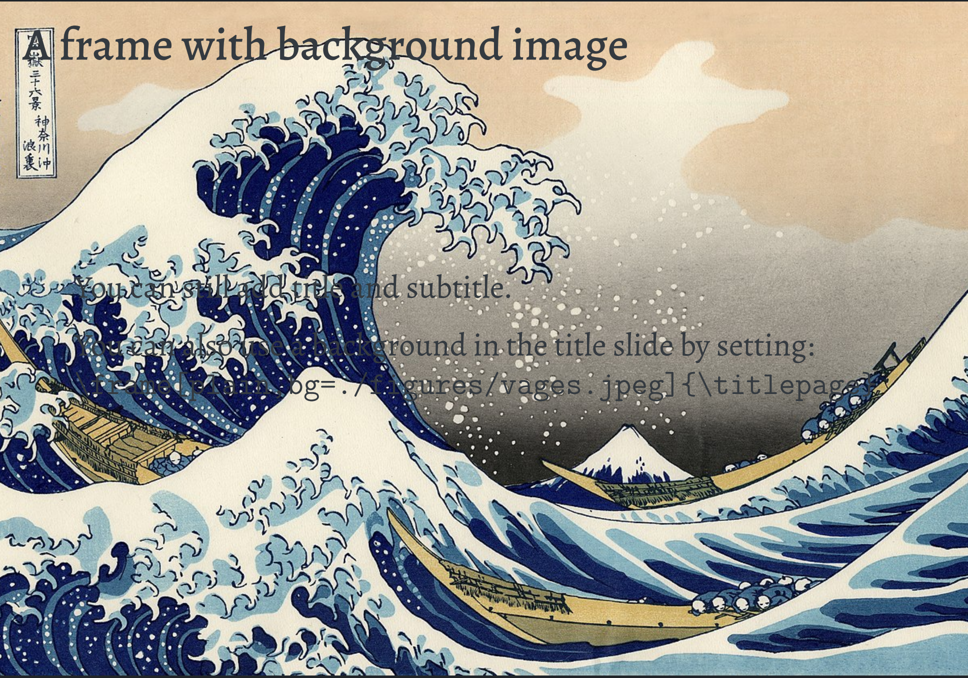
Démonstration

$$e^{iz} = \cos z + i \sin z$$

therefore

$$\begin{aligned} e^{i\pi} + 1 &= \cos \pi + i \sin \pi + 1 \\ &= -1 + i \times 0 + 1 \\ &= 0 \end{aligned}$$

A frame with background image



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A ***STANDOUT*** frame can be used to focus attention

In combination with *plain*,
it makes a nice thank-you slide!

https://github.com/piazzai/arguelles
https://ctan.org/pkg/beamertheme-arguelles